

Multiple Regression

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Quantitative Reasoning II · INF 1220 · University of Austin

Lesson 9

Last Time

Simple linear regression on 2,925 home sales in Ames, Iowa

Predictor	Slope	R^2
Living area, sq ft	\$116.23	0.518
Lot area, sq ft	\$2.73	0.073
Age at sale, years	-\$1,467	0.320
Garage capacity, cars	\$67,473	0.426
Quality rating, 1–10	\$45,087	0.648

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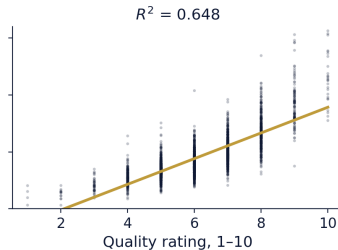
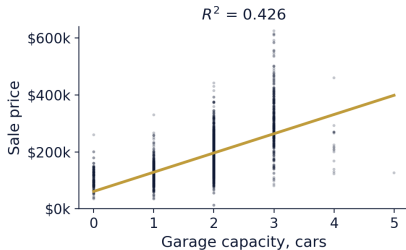
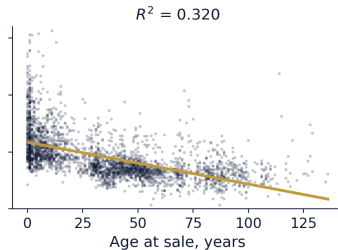
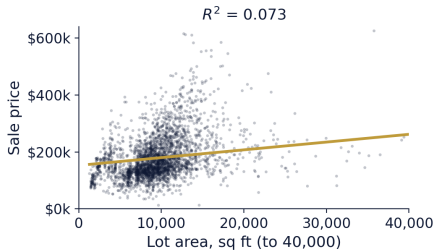
Today we'll add more predictors to our model

Same Size, Different Price

Neighbourhood	Area	Lot area	Age	Quality	Garage	Sold for
Old Town	1,440	5,000	106	5	1	\$79,000
Edwards	1,440	7,711	30	4	0	\$150,000
Timberland	1,442	9,802	2	5	2	\$172,500
College Creek	1,431	9,017	10	7	2	\$192,000
Sawyer West	1,433	11,980	23	7	2	\$270,000

Holding area fixed, we see that other characteristics may be correlated with price.

Price and the Other Predictors



Multiple Linear Regression Model

With more than one independent variable, our regression model becomes:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + e$$

- Y : the dependent variable, sale price
- $X_1, \dots, X_p$: the p independent variables
- β_0 : the intercept
- $\beta_1, \dots, \beta_p$: the slopes, one for each independent variable.
- e : the residual, $Y - \hat{Y}$

Interpreting a Slope

Each slope is the average change in Y for a one-unit increase in its X , holding all the other X 's constant.

For us:

Y = price, X_1 = living area, X_2 = quality rating

β_1 compares houses of the *same quality* that differ in size.

β_2 compares houses of the *same size* that differ in quality.

Least Squares, Again

The rule is the same as last time.

- Fitted value: $\hat{Y}_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i}$
- Residual: $e_i = Y_i - \hat{Y}_i$
- Choose β_0 , β_1 , and β_2 to minimise the sum of squared residuals, $SSE = \sum e_i^2$

R^2 is the proportion of the variation in Y that the model explains.

How R^2 Is Calculated

$$SST = \sum (Y_i - \bar{Y})^2$$

$$SSE = \sum e_i^2$$

$$R^2 = 1 - \frac{SSE}{SST}$$

- SST: the squared error from predicting $\bar{Y}$ for every house
- SSE: the squared error left after the model
- R^2 : the share of SST the model removes

Adding a predictor never lowers R^2 , and often increases it, even if the predictor has low explanatory value.

Price on Area and Quality

$$\widehat{\text{price}} = -113,057 + 62.78 \times \text{area} + 32,797 \times \text{quality}$$

Area: holding quality constant, each additional square foot is associated with a \$62.78 higher price, on average.

Quality: holding area constant, each additional point of quality is associated with a \$32,797 higher price, on average.

Intercept: a house with zero square feet and a quality rating of zero. Not useful on its own.

Comparing Models

	Area only	Quality only	Both
Living area	\$116.23		\$62.78
Quality		\$45,087	\$32,797
R^2	0.518	0.648	0.751

Both slopes are smaller in the model with both predictors

R^2 is not the sum

Why Did the Slopes Fall?

- $\text{corr}(\text{area}, \text{quality}) = 0.56$: bigger houses tend to be better built
- Area alone: part of the \$116.23 is quality
- Holding quality constant: \$62.78
- Quality is a **lurking variable** in the area-only model: related to both area and price, and left out

Adding Lot Area or Age

	Area only	Area and lot area	Area and age
Living area	\$116.23	\$112.46	\$100.07
Lot area		\$0.91	
Age at sale			−\$1,081
R^2	0.518	0.525	0.681

Lot area and age are about equally correlated with area: 0.26 and −0.24

Categorical Variables: Binary Variables

Not every variable is a number!

Does the house have central air conditioning?

$$\text{air} = \begin{cases} 1 & \text{if the house has central air} \\ 0 & \text{if it does not} \end{cases}$$

A 0/1 variable is called a **dummy** or a **indicator**, or a **binary variable**.
In our data, 2,729 houses have central air and 196 do not.

Regression on a Dummy Variable

$$\widehat{\text{price}} = \beta_0 + \beta_1 \times \text{air}$$

One model for each group:

$$\text{No central air, air} = 0 \quad \widehat{\text{price}} = \beta_0 + \beta_1 \times 0 = \beta_0$$

$$\text{Central air, air} = 1 \quad \widehat{\text{price}} = \beta_0 + \beta_1 \times 1 = \beta_0 + \beta_1$$

β_0 is the mean price of houses without central air. β_1 is the difference between the two group means.

Central Air Only

$$\widehat{\text{price}} = 101,890 + 84,161 \times \text{air}$$

No central air	$\widehat{\text{price}} = \$101,890$
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Central air	$\widehat{\text{price}} = 101,890 + 84,161 = \$186,051$
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Houses with central air sold for \$84,161 more, on average.

Adding Living Area

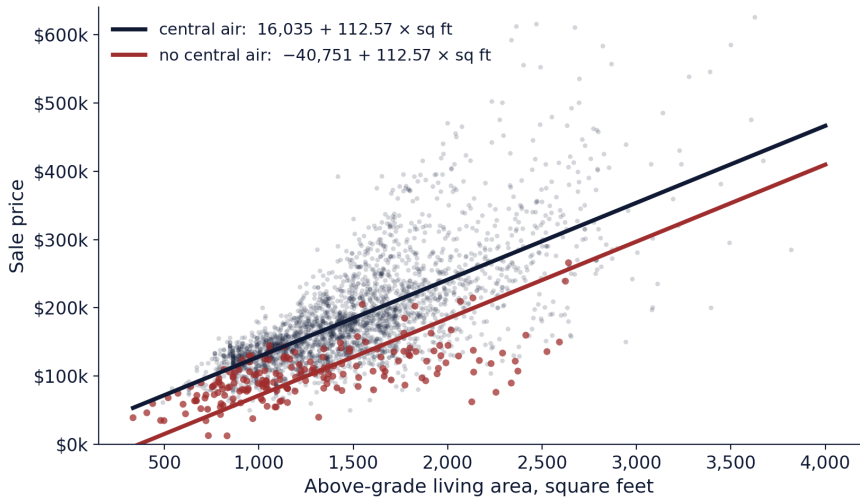
$$\widehat{\text{price}} = -40,751 + 112.57 \times \text{area} + 56,785 \times \text{air}$$

For each group:

No central air	$\widehat{\text{price}} = -40,751 + 112.57 \times \text{area}$
Central air	$\widehat{\text{price}} = 16,035 + 112.57 \times \text{area}$

Same slope, different intercepts.

Two Parallel Lines



Interpreting the Dummy

Among houses of the same size, houses with central air sold for \$56,785 more, on average.

$$16,035 - (-40,751) = 56,785.$$

More Than Two Categories

Let's consider three different neighbourhoods:

	Sales	nbhd2	nbhd3
Neighbourhood 1	443	0	0
Neighbourhood 2	239	1	0
Neighbourhood 3	166	0	1

We have three categories, so we need two indicators.

What would happen if we had three indicators?

The category with no dummy is the **base category**: here, neighbourhood 1.

Each indicator's coefficient is a comparison with the base category.

One Model for Each Neighbourhood

$$\widehat{\text{price}} = \beta_0 + \beta_1 \times \text{nbhd2} + \beta_2 \times \text{nbhd3}$$

	nbhd2	nbhd3	Model
Neighbourhood 1	0	0	$\widehat{\text{price}} = \beta_0$
Neighbourhood 2	1	0	$\widehat{\text{price}} = \beta_0 + \beta_1$
Neighbourhood 3	0	1	$\widehat{\text{price}} = \beta_0 + \beta_2$

β_0 is the mean price in neighbourhood 1. β_1 and β_2 are each neighbourhood's difference from neighbourhood 1.

Neighbourhood Only

$$\widehat{\text{price}} = 145,097 - 21,105 \times \text{nbhd2} + 176,921 \times \text{nbhd3}$$

Neighbourhood 1	$\widehat{\text{price}} = \$145,097$
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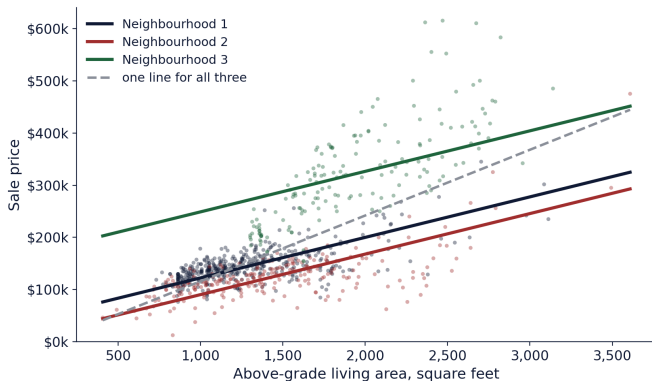
Neighbourhood 2	$\widehat{\text{price}} = 145,097 - 21,105 = \$123,992$
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Neighbourhood 3	$\widehat{\text{price}} = 145,097 + 176,921 = \$322,018$
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Neighbourhood 2 houses sold for \$21,105 less than neighbourhood 1 houses, on average.

Adding Living Area

$$\widehat{\text{price}} = 44,845 + 77.59 \times \text{area} - 31,962 \times \text{nbhd2} + 126,441 \times \text{nbhd3}$$



Neighbourhood Gaps With and Without Living Area

	Without area	With area
Neighbourhood 2	−\$21,105	−\$31,962
Neighbourhood 3	\$176,921	\$126,441
Living area		\$77.59

Changing the Base Category

With neighbourhood 2 as the base instead:

	Base: neighbourhood 1	Base: neighbourhood 2
Neighbourhood 1		\$31,962
Neighbourhood 2	-\$31,962	
Neighbourhood 3	\$126,441	\$158,403
Living area	\$77.59	\$77.59

The coefficients change because they compare with a different group. Every fitted value is the same.

Reading Regression Output

Price on four predictors. $n = 2,925$. $R^2 = 0.775$.

	Coefficient	Std. error	95% CI	p-value
Intercept	-60,676	5,238	(-70,948, -50,405)	< 0.001
Living area, sq ft	67.29	1.74	(63.89, 70.70)	< 0.001
Quality, 1-10	25,317	730	(23,885, 26,748)	< 0.001
Age at sale, years	-493	30	(-551, -435)	< 0.001
Central air, 0/1	4,738	2,979	(-1,103, 10,579)	0.11